25900287 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSECHE201 - INTRODUCTION TO NANOSCIENCE

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **5** questions

Each question carries **4** marks

1. How do geckos' feet exhibit nanoscale features that enable them to stick to surfaces? [U] / [CO1]
2. List four applications of TEM. [K] / [CO2]
3. Explain the concept of quantum dots with an example. [U] / [CO3]
4. How does nanotechnology contribute to solar energy harvesting? [U] / [CO4]
5. Explain the characteristics and potential applications of one-dimensional nanoparticle [K] / [CO1]
6. Recall what carbon dots are. Mention one method of their synthesis and list the raw materials used as carbon sources in their synthesis. [K] / [CO3]
7. How do nanomaterials facilitate data storage? [An] / [CO4]

[4x5 = 20]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **7** marks

8. Describe how magnetic nanoparticles help in enviornmental remediation. [U] / [CO3]

9. Discuss the role of nanotechnology in developing biosensors. Illustrate with examples. [An] / [CO4]

10. Discuss the advantages of nanotechnology-based diagnosis over conventional methods. [U] / [CO4]

11. Write the working principle and procedure of chemical precipitation method for nanomaterial synthesis. How pH, temperature and precursor concentration influence particle size and morphology of synthesised nanomaterials? [U] / [CO1]

12. Discuss the principle of AFM. Evaluate its importance in the characterization of nanomaterials. [U] / [CO2]

13. Discuss Richard Feynman's hypothesis on atomic-scale manipulation and explain its historical significance, and how it laid the foundation for the development of nanotechnology. [U] / [CO1]

14. Explain the importance of UV - Visible spectra and IR spectra in the characterization of nanomaterials [U] / [CO2]

[7x5 = 35]

Part C

Essay Type Questions

Answer any **1** questions

Each question carries **15** marks

15. Compare SEM and TEM [U] / [CO2]

16. Explain the top down approaches used in nanochemistry. Discuss its methods, principle, advantages and limitations with suitable examples. [U] / [CO1]

[15x1 = 15]